Data Management Plan

1. Type of data generated

This dataset was generated from plant-pollinator surveys conducted at two restoration sites within Arthur's Seat, Edinburgh. The data consisted primarily of quantitative records of pollinator visitation, floral abundance, plant identity, survey timing, and environmental conditions. Pollinator visitation was recorded during 20-minute focal plant surveys, with flower visitors identified to the lowest reliable taxonomic level. Environmental data collected during each survey included temperature (°C), solar radiation ($W\ m^{-2}$), relative humidity (% RH), and wind speed ($m\ s^{-1}$). Floral abundance was quantified using direct counts or estimates from 1 x 1 m quadrats depending on the abundance and growth form of each plant species. The data were subsequently used for statistical analyses of pollinator visitation and community composition and to construct quantitative plant-pollinator interaction networks. Final outputs include summary tables, statistical model outputs, figures, interaction matrices, network visualisations, and network matrices including connectance, weighted nestedness and modularity.

2. Types of data preserved

The raw pollinator survey dataset, cleaned datasets used for statistical and network analyses, and associated metadata will be preserved. All R scripts used to clean, analyse, and visualise the data will also be stored in a GitHub repository, allowing the analyses presented within the dissertation to be reproduced. Supporting information required to interpret the dataset, including plant identities, pollinator classifications, floral abundance estimates, and descriptions of recorded variables, will be retained alongside the data.

3. Software and metadata implications

Datasets will be stored in commonly accessible formats, principally .csv and .xlsx, which can be opened using spreadsheet software such as Microsoft Excel or LibreOffice Calc. All statistical analyses and data visualisations were conducted in R, with the R scripts retained alongside the datasets to facilitate reproducibility. Metadata will accompany the dataset to describe variable names, units, sampling structure, plant identities, pollinator taxonomic classifications, and the methods used to quantify floral abundance. Scientific names and taxonomic classifications will be recorded consistently throughout the archived data and associated documentation.

4. Length of data preservation

The full dataset, including the field notebook images, raw survey data, metadata, and R code, will be deposited in the associated GitHub repository, where it will remain available following completion of the dissertation.

5. Value of data to others

The dataset provides information on plant-pollinator interactions within restored plant communities on Arthurs Seat and may therefore provide a helpful baseline for future monitoring of the restoration sites. In particular, the interaction records and floral abundance data could allow future surveys to assess changes in pollinator visitation, plant-pollinator network structure, and the integration of restored plant populations into the wider ecological community. Preservation of the data may also be valuable to researchers and land managers interested in urban biodiversity, ecological restoration, pollinator communities, and plant-pollinator interaction networks.

6. Length of proprietary period

There is no proprietary period associated with this dataset, as the data and associated analytical code have been made publicly available through the project GitHub repository.

7. How data will be shared

The field notebook images, raw dataset, metadata, and R scripts will be shared through the GitHub repository listed in the dissertation's data availability statement. Data will be provided in accessible formats, including .csv and .xlsx, with analytical code provided as .R files. Documentation describing variable definitions, units, plant identities, pollinator classifications, and relevant methodological information will accompany the data to facilitate interpretation and reuse.

8. Resources needed to preserve and share data

No specialised resources are required beyond existing software and online storage. GitHub will be used for data and code sharing and version control, while R will be used for data processing, statistical analysis, and visualisation. Spreadsheet data will be stored in widely accessible .csv and .xlsx formats, allowing the archived material to be accessed without specialist infrastructure.